a Whose knowledge transfers?
Chemists reuse molecular experience
Amine-based ligand guides[2]
[M] / L
Ar–X + H–Nu
Ar–Nu
General conditions[6–9]
base / solvent
Molecules meet new partners
Informer libraries[3,4]
NH2
O
Br
Robustness screens[5]
63% recur with partners
Molecules with >1 partner
Which molecule carries the conditions?
Never tested
b Replace either partner
Every pair, same conditions
Molecule B
A
Each pair
conditions
177,039 reactions
15 screens · 14 transformations
Fix A; replace B
Fix B; replace A
Molecular vs pair-specific
A
B
Pair-specific
Molecular share
844 outside-screen pairs
2 external public screens
c One molecule dominates
Median 80% to one molecule
Owner / molecular component
Roles switch with reaction
Amide coupling
Owner
C–N coupling
Partner
4-methylaniline
10/10 halide screens
halide
amine / N–H
other
Order: Fig. 2a
≈3× condition-profile prediction
Preference explained
Owner
Partner
Set A · Fig. 5a